

Qasim Sethar

AI Systems Engineer · Autonomous Agents · Retrieval Infrastructure · Applied ML

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Professional Summary

AI Systems Engineer with hands-on experience architecting autonomous agent loops, production-grade retrieval-augmented generation (RAG) pipelines, and high-performance developer tooling. Built **Engram** — a local-first personal knowledge OS utilizing hybrid sparse/dense retrieval with strict grounded constraints to eliminate hallucinations. Developed **Operon** — a terminal-native autonomous coding agent running a persistent cross-file symbol graph and deterministic AST verification layer. Published **foldr** on PyPI — a cross-platform file automation CLI tracking 5,000+ downloads. Strong foundation in classical information retrieval; designs deterministic software systems where data safety, tenant isolation, and architectural correctness are first-class constraints.

Technical Skills

Languages: Python (4 yrs), Java (2 yrs), C++ (1.5 yrs), SQL, Bash

Agentic AI: ReAct Architecture, Tool Calling, Multi-step Planning, Self-correction, Human-in-the-Loop Oversight

LLM & ML: LlamaIndex, LangChain, HuggingFace Transformers, PyTorch, BERT Fine-tuning, Cross-Encoders

Retrieval: RAG Pipelines, Hybrid Search (Sparse+Dense), Reciprocal Rank Fusion (RRF), pgvector, LanceDB, ChromaDB

Parsing / AST: Python `ast`, `tokenize`, Tesseract OCR, Poppler, Document Intelligence, BeautifulSoup4

Systems: Linux (Arch/Ubuntu), Docker, Redis, Celery, GitHub Actions, Daemon Processes, OS Filesystem APIs, PyPI Packaging

Tooling: Next.js, pnpm, TanStack Query, FastAPI, SQLAlchemy, Pydantic, PostgreSQL, Cloudflare R2, Git

Engineering Projects

Engram — Local-First Personal Knowledge OS

Next.js · FastAPI · PostgreSQL (pgvector) · Celery · Redis · RAG

<https://github.com/qasimio/engram>

- Architected a local-first personal knowledge engine with workspace-first multi-tenancy, enforcing strict row-level data isolation via FastAPI dependency injection to guarantee cross-tenant privacy at the core architectural layer.
- Engineered a high-precision hybrid retrieval pipeline fusing dense vector search (pgvector cosine) and sparse BM25 keyword search using Reciprocal Rank Fusion (RRF), coupled with a Cross-Encoder reranker to solve exact-match failures.
- Designed an asynchronous ingestion system powered by a Redis-backed Celery worker pool, decoupling multi-format file parsing, web scraping, and automated GitHub repository syncing from the main API request thread.
- Implemented a hard-boundary grounded synthesis engine that mandates inline source citations and forces the LLM to return an explicit "not found" status rather than fallback to training-data hallucinations when context evidence is absent.
- Built an abstract LLM routing layer supporting 7 hot-swappable local/cloud backends (Ollama, llama.cpp, OpenAI, Anthropic) alongside an MCP integration layer to expose knowledge bases securely to desktop agent ecosystems.

Operon — Autonomous Code Intelligence Agent

Python · Textual · LLM APIs · AST · ReAct

<https://github.com/qasimio/Operon>

- Architected a terminal-native autonomous agent constructing a persistent, hash-gated repository symbol graph for cross-file refactoring, execution flow mapping, and contextual analysis without overloading context windows.
- Engineered a deterministic reviewer that validates file edits against disk content snapshots prior to LLM interaction, eliminating hallucinated change confirmations—the primary failure point of local autonomous loops.
- Built a structured CRUD fast-path using Python `ast` and `tokenize` modules for predictable codebase edits (such as import insertions and symbol renaming), dropping LLM-induced syntax errors to zero for these operations.
- Designed a 5-tier surgical diff engine executing cascading matching from exact string to fuzzy multi-line structures, ensuring highly reliable patch application across overlapping automated and manual edits.
- Developed a hot-swappable LLM router supporting 9 unique backends with immediate runtime evaluation transitions, complemented by a mandatory human-approval gate layout with automatic 300s fallback rejections.

MQNotebook — Enterprise RAG System

Python · LlamaIndex · Tesseract · Poppler · Streamlit

<https://github.com/qasimio/MQNotebook>

- Engineered a local-first RAG tool targeting complex enterprise document formats including scanned PDFs, PPTX speaker notes, and multi-sheet XLSX formats via a custom-wrapped Tesseract and Poppler OCR extraction layer.
- Implemented hybrid retrieval methodologies backed by Cross-Encoder reranking models, generating a **~40% precision increase** in semantic context matching accuracy compared to baseline naive cosine similarities.
- Neutralized persistent WinError 32 file-locking deadlocks within target vector stores on Windows systems by building a session-isolated storage supervisor that routes simultaneous lookups to separate persistent runtime handlers.
- Optimized context payload compression strategies, **minimizing token overhead by ~60%** while maintaining document retrieval integrity under tight target container deployment limits.

DevShelf — Vertical Search Engine (from First Principles)

Java · Information Retrieval · Data Structures

<https://github.com/qasimio/DevShelf>

- Architected a custom vertical search engine for scientific literature entirely from scratch without external indexing engines, designing a high-performance Positional Inverted Index ensuring **O(1)** keyword retrieval.
- Engineered a decoupled offline indexing pipeline that isolates background index generation from foreground runtimes, securing sub-millisecond query execution latencies along with Trie-based prefix matching.

FOLDER — File Automation CLI `pip install foldr`

Python · PyPI · watchdog · Daemon Processes · Cross-platform

<https://github.com/qasimio/foldr>

- Designed and published a cross-platform file automation CLI to PyPI mapping **240+ extensions** into **30+ logical categories**, accumulating over 5,000 downloads for automated pipeline data staging tasks.
- Engineered an efficient, low-overhead background daemon using native OS filesystem events—inotify (Linux), kqueue (macOS), and ReadDirectoryChangesW (Windows) via the `watchdog` engine—consuming 0% CPU when idle.
- Built a transaction-safe JSON undo tracker logging historical I/O states, allowing independent non-sequential rollback operations using git-revert style structural file mapping logic.
- Designed a robust preview engine displaying planned I/O passes prior to execution, integrating automatic duplicate name resolution schemes and protected path blocks to eliminate unintended source modifications.

Experience

Machine Learning Intern · Arch Technologies (Remote)

Present

- Fine-tuned BERT models for production NLP text classification workflows; profiled execution bottlenecks to optimize training preprocessing sequences, securing measurable gains in pipeline step throughput.
- Integrated fine-tuned PyTorch weights into internal software prototypes, handling model serialization, version tracking, and containerized local model inference deployment architectures.

Open Source & Published Tools

- **DevShelf** · Search Engine — High-performance first-principles offline search engine for technical documents written in Java. qasimio.me/projects/devshelf
- **Operon** · GitHub — Open-source terminal-native autonomous coding agent with deterministic AST verification. github.com/qasimio/Operon
- **FOLDER v2.1** · PyPI — Cross-platform background file orchestration utility with active state monitoring. pypi.org/project/foldr

Additional Technical Work

BabyGPT — Character-level LSTM language model implemented from scratch in TensorFlow to validate deep sequence modeling foundations.

MQ Banking Core — Transactional C++ banking structure enforcing local file I/O operations and rigid balance integrity guarantees. **Digital Eye**

— Convolutional Neural Network (CNN) computer vision classification pipeline built using Keras and TensorFlow.

Education

B.Sc. Computer Science (5th Semester Student) · Sukkur IBA University, Pakistan

Expected 2028

- Focus Areas: Distributed Computing, Architecture of AI Systems, Advanced Information Retrieval, Data Structures & Algorithms.